



GOSFORD HIGH SCHOOL

**2007
YEAR 12 HALF YEARLY HSC COURSE.**

MATHEMATICS EXTENSION 1

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Each question should be started on a new page.
- All necessary working should be shown in every question

Total marks: - 84

- Attempt Questions 1 -7
- All questions are of equal value

Question 1 (12 marks)

(i) Find the acute angle between the lines $2y - x + 1 = 0$ and $5x - y + 2 = 0$ correct to the nearest minute. (2)

(ii) Find the coordinates of the point P that divides the interval joining A (2,5) and B (-1,0) externally in the ratio 3:4. (2)

(iii) Solve $\frac{x+1}{x-1} \geq 3, x \neq 1$ and graph your solution on a number line. (3)

(iv) Differentiate with respect to x:

(a) $x^2 \cos^{-1} x$. (2)

(b) $\frac{\tan^{-1} x}{x}$. (3)

Question 2 (12marks)

(i) Find in simplest surd form the exact value of $\sin 75^\circ$. (3)

(ii) If α, β & γ are the roots of the equation $2x^3 + 5x - 3 = 0$ find the value of

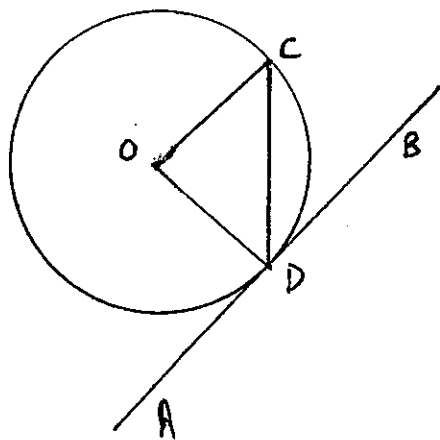
(a) $\alpha + \beta + \gamma$ (1)

(b) $\alpha\beta\gamma$ (1)

(c) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ (2)

(iii) Draw a neat sketch of $y = 2 \sin^{-1} \frac{x}{3}$ and state its domain and range. (3)

(iii) AB is a tangent at D to the circle center O. Calculate the size of $\angle CDB$ given that $\angle COD = 110^\circ$ (2)



NOT TO SCALE

Question 3 (12 marks)

(i) Find

$$(a) \int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} \quad (2)$$

$$(b) \int \sin^2 x \, dx \quad (2)$$

$$(c) \int_0^{\frac{\pi}{2}} \cos^2 2x \, dx \quad (2)$$

$$(d) \int_1^2 2x\sqrt{x^2-1} \, dx \text{ using the substitution } u=x^2-1 \quad (3)$$

(ii) Use the substitution $u = \sin x$ to show that:

$$\int_0^{\frac{\pi}{6}} \frac{\cos x \, dx}{4\sin^2 x + 1} = \frac{\pi}{8} \quad (3)$$

Question 4 (12 marks)

(i) Express $\tan 2A$ in terms of $\tan A$ and without using a calculator show that $\tan^{-1}\left(\frac{12}{5}\right) = 2 \tan^{-1}\left(\frac{2}{3}\right)$. (3)

(ii) (a) Write $\sqrt{3} \sin x - \cos x$ in the form $R \sin(x - \alpha)$ where $R > 0$ & $0 \leq \alpha \leq \frac{\pi}{2}$. (1)

(b) Hence or otherwise solve $\sqrt{3} \sin x - \cos x = 1$ for $0 \leq x \leq 2\pi$ (2)

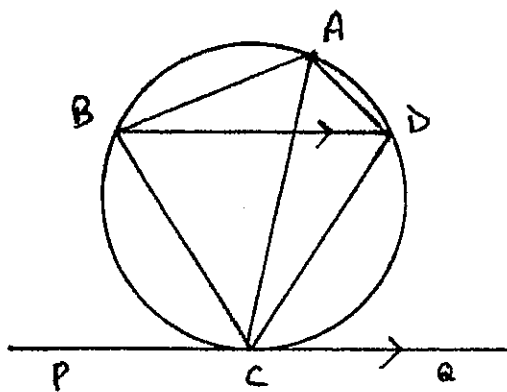
(iii) Prove by the principle of mathematical induction that:

$$1 + 4 + 4^2 + \dots + 4^{n-1} = \frac{1}{3}(4^n - 1) \quad (4)$$

(iv) If $y = \sin^{-1} \sqrt{x}$, show that $\frac{dy}{dx} = \frac{1}{\sin 2y}$ (2)

Question 5 (12 marks)

(i) Copy the following diagram onto your answer sheet.



NOT TO SCALE

Given that BD is parallel to PQ and PQ is a tangent to the circle at C, prove that CA bisects $\angle BAD$. (4)

(ii) (a) Show that the equation $\log_e x - \cos x = 0$ has a root between $x=1$ and $x=2$. (1)

(b) Using one application of Newton's Method find a value of the root of the equation $\log_e x - \cos x = 0$, which is close to $x=1.2$. (3)

(iii) The normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$ cuts the axis of the parabola at Q. If S is the focus of the parabola, find the area of triangle PQS. (4)

Question 6 (12 marks)

(i) Find all solutions to $\sin 2\theta = \cos \theta$ (3)

(ii) (a) Show that $\frac{d}{dx} [x \sin^{-1} x + \sqrt{1-x^2}] = \sin^{-1} x$ (2)

(b) Hence show that the volume of the solid of revolution formed by rotating the area under the curve $y = \sqrt{\sin^{-1} x}$ between $x=0$ & $x=\frac{1}{2}$, about the x-axis, is $\frac{\pi^2}{12} + (\sqrt{3}-2)\frac{\pi}{2}$ units³. (3)

(iii) (a) On the same set of axes sketch the graphs of $y=2|x|$ & $y=|x+3|$ (2)

(b) Solve $2|x| \leq |x+3|$ (2)

Question 7 (12 marks)

(i) Find $\tan\left[\sin^{-1}\frac{5}{13}-\sin^{-1}\frac{4}{5}\right]$ (2)

(ii) $P(4t, 2t^2)$ and $Q(8t, 8t^2)$ are two variable points on the parabola $x^2 = 8y$. The tangents from P and Q intersect at T . Find the Cartesian equation of the locus of T . (4)

(iii) (a) Sketch the graph of the function $f(x) = e^x - 4$ clearly showing the coordinates of any points of intersection with the axes and the equations of any asymptotes. (2)

(b) On the same diagram sketch the graph of the function $y = f^{-1}(x)$ showing all important features. (1)

(c) Find an expression for $y = f^{-1}(x)$ in terms of x . (2)

(d) Explain why the x coordinates of any points of intersection of the two functions satisfy the equation $e^x - x - 4 = 0$. (1)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1.

(i) $M_1 = \frac{1}{2}, M_2 = 5$

$$\tan \theta = \frac{|M_1 - M_2|}{|1 + M_1 M_2|}$$

$$= \left| \frac{\frac{1}{2} - 5}{1 + \frac{1}{2} \times 5} \right|$$

$$= \frac{4\frac{1}{2}}{3\frac{1}{2}}$$

(2)

$$\therefore \theta = 52^\circ 8' \text{ (nearest min)}$$

(ii) $(2, 5) \quad (-1, 0)$

$$3 : -4$$

$$x = \frac{-4 \times 2 + 3 \times -1}{3 + -4}, \quad y = \frac{-4 \times 5 + 3 \times 0}{3 + -4}$$

$$= 11$$

$$= 20$$

(2)

P is the pt $(11, 20)$

(iii) $\frac{x+1}{x-1} > 3$

$$x \neq 1$$

Consider the eqⁿ:

$$\frac{x+1}{x-1} = 3$$

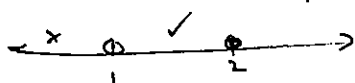
$$x+1 = 3x-3$$

$$4 = 2x$$

$$x = 2$$

Since this is the solⁿ to the eqⁿ it can't be a solⁿ to the ineqⁿ.

(3)

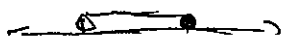


Test 0: $\frac{1}{-1} \not> 3$

Test $\frac{1}{2}$: $\frac{2\frac{1}{2}}{\frac{1}{2}} > 3$

Test 3: $\frac{4}{2} \not> 3$

$$\therefore \text{Solⁿ is } 1 < x \leq 2$$



(iv) c) Let $y = x^2 \cos^{-1} x$

$$\frac{dy}{dx} = SF' + FS'$$

(2)

$$= \cos^{-1} x \cdot 2x + x^2 \cdot \frac{-1}{\sqrt{1-x^2}}$$

$$= 2x \cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}}$$

b) Let $y = \frac{\tan^{-1} x}{x}$

$$\frac{dy}{dx} = \frac{BT' - TB'}{B^2}$$

(3)

$$= x \cdot \frac{1}{1+x^2} - \tan^{-1} x \cdot 1$$

$$= \frac{\frac{x}{1+x^2} - \tan^{-1} x}{x^2}$$

$$= \frac{x - (1+x^2)\tan^{-1} x}{x^2(1+x^2)}$$

Question 2

(i) $\sin 75^\circ = \sin(45^\circ + 30^\circ)$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \quad (3)$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

(ii) a) $\alpha + \beta + \gamma = -\frac{b}{a}$
 $= 0$ (1)

b) $\alpha\beta\gamma = -\frac{c}{a}$
 $= \frac{3}{2}$ (1)

c) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$
 $= \frac{\frac{c}{a}}{\frac{3}{2}}$
 $= \frac{\frac{5}{2}}{\frac{3}{2}}$ (2)
 $= \frac{5}{3}$

(iii) If $y = 2 \sin^{-1} \frac{x}{3}$

$$\frac{y}{2} = \sin^{-1} \frac{x}{3}$$

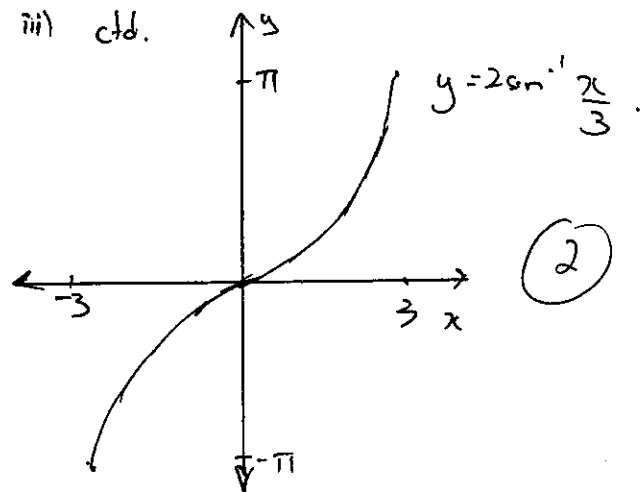
$$\therefore R: -\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$$

$$\text{i.e. } -\pi \leq y \leq \pi$$
 (1)

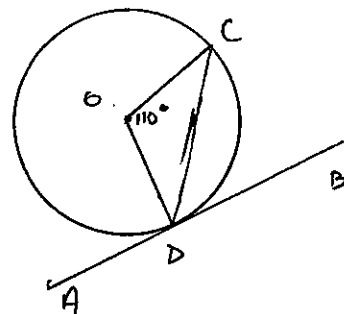
a) D: $-1 \leq \frac{x}{3} \leq 1$

$$-3 \leq x \leq 3$$

iii) ctd.



(iv)



$\angle OCD = \angle ODC = 35^\circ$ (L sin of a Δ is 180° & base Ls of an isosc Δ)

$\therefore \angle CDB = 35^\circ$ (L in the alt segment theorem). (2)

Question 3 :

$$\begin{aligned} \text{(i) a) } \int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} \\ &= \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} \quad (2) \\ &= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{3} - 0 \\ &= \frac{\pi}{3}. \end{aligned}$$

$$\begin{aligned} \text{b) } \int \sin^2 x \, dx \\ &= \int \frac{1}{2} - \frac{1}{2} \cos 2x \, dx \quad (2) \\ &= \frac{x}{2} - \frac{1}{4} \sin 2x + c \end{aligned}$$

$$\begin{aligned} \text{c) } \int_0^{\frac{\pi}{2}} \cos^2 x \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 2x \, dx \\ &= \left[\frac{x}{2} + \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{2}} \quad (2) \\ &= \left(\frac{\pi}{4} + \frac{1}{4} \sin \pi \right) - \left(\frac{0}{2} + \frac{1}{4} \sin 0 \right) \\ &= \left(\frac{\pi}{4} + 0 \right) - (0 + 0) \\ &= \frac{\pi}{4}. \end{aligned}$$

$$\begin{aligned} \text{d) } \text{If } u &= x^2 - 1 \\ du &= 2x \, dx \\ \text{When } x &= 1, \quad u = 0, \\ x &= 2, \quad u = 3. \end{aligned}$$

d) c+d.

$$\begin{aligned} \therefore I &= \int_0^3 u^{\frac{1}{2}} \, du \\ &= \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^3 \quad (3) \\ &= \frac{2}{3} \times 3^{\frac{3}{2}} - \frac{2}{3} \times 0 \\ &= \frac{2}{3} \times 3\sqrt{3} \\ &= 2\sqrt{3}. \end{aligned}$$

$$\begin{aligned} \text{(ii) If } u &= \sin x \\ du &= \cos x \, dx \end{aligned}$$

$$\begin{aligned} \text{When } x &= 0, \quad u = 0 \\ x &= \frac{\pi}{6}, \quad u = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \therefore I &= \int_0^{\frac{1}{2}} \frac{du}{u^2 + 1} \quad (3) \\ &= \frac{1}{4} \int_0^{\frac{1}{2}} \frac{du}{u^2 + \frac{1}{4}} \\ &= \frac{1}{4} \times 2 \left[\tan^{-1} 2u \right]_0^{\frac{1}{2}} \\ &= \frac{1}{2} \left[\tan^{-1} 1 - \tan^{-1} 0 \right] \\ &= \frac{1}{2} \times \left(\frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{8}. \end{aligned}$$

Question 4:

$$(i) \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\text{Let } A = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \tan A = \frac{2}{3}$$

$$\begin{aligned} \tan 2A &= \frac{2 \times \frac{2}{3}}{1 - \left(\frac{2}{3}\right)^2} \quad (3) \\ &= \frac{\frac{4}{3}}{\frac{5}{9}} \\ &= \frac{4}{3} \times \frac{9}{5} \\ &= \frac{12}{5} \end{aligned}$$

$$\therefore 2A = \tan^{-1}\left(\frac{12}{5}\right)$$

$$\text{i.e. } 2 \tan^{-1}\left(\frac{2}{3}\right) = \tan^{-1}\left(\frac{12}{5}\right)$$

$$(ii) \text{ a) } R = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\text{a) } \tan \alpha = \frac{1}{\sqrt{3}} \quad (1)$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \sin x - \cos x = 2 \sin\left(x - \frac{\pi}{6}\right)$$

$$\begin{aligned} \text{b) } \sqrt{3} \sin x - \cos x &= 1 \\ 2 \sin\left(x - \frac{\pi}{6}\right) &= 1 \\ \sin\left(x - \frac{\pi}{6}\right) &= \frac{1}{2} \quad (2) \\ \therefore x - \frac{\pi}{6} &= \frac{\pi}{6}, \frac{5\pi}{6} \\ x &= \frac{\pi}{3}, \pi \end{aligned}$$

(iii) Assume true for $n=k$

$$\text{i.e. } 1 + 4 + 4^2 + \dots + 4^{k-1} = \frac{1}{3}(4^k - 1)$$

Prove true for $n=1$

$$\text{LHS} = 1$$

$$\text{RHS} = \frac{1}{3}(4^1 - 1)$$

$$= 1$$

$\therefore$ true for $n=1$.

Prove true for $n=k$

$$\text{i.e. } 1 + 4 + \dots + 4^{k-1} + 4^k = \frac{1}{3}(4^{k+1} - 1)$$

$$\begin{aligned} \text{LHS} &= \frac{1}{3}(4^k - 1) + 4^k \\ &= \frac{(4^k - 1) + 3 \times 4^k}{3} \quad (4) \\ &= \frac{4 \times 4^k - 1}{3} \\ &= \frac{1}{3}(4^{k+1} - 1) \end{aligned}$$

$\therefore$ If the statement is true for $n=k$ it is true for $n=k+1$.
Since it is true for $n=1$ it is true for $n=2$ & so on.
 $\therefore$ the statement is true for all positive integers n .

$$(iii) \text{ I f } y = \sin^{-1} \sqrt{x}$$

$$\sqrt{x} = \sin y$$

$$x = \sin^2 y$$

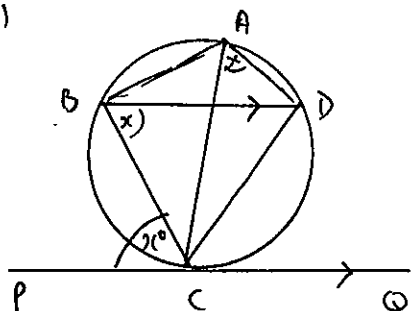
$$\therefore \frac{dx}{dy} = 2 \sin y \cos y \quad (2)$$

$$= \sin 2y$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sin 2y}$$

Question 5

(i)



Let $\angle PCB$ be x°

$\therefore \angle CBD = x^\circ$ (alt. L's in parallel lines, $BD \parallel PQ$)

$\therefore \angle DAC = x^\circ$ (Angles in the same segment are equal)

Also $\angle BAC = x^\circ$ (L is the alt. segment theorem)

$$\therefore \angle DAC = \angle BAC$$

ie. CA bisects $\angle BAD$.

(ii) a) $\log_e | - \cos | \doteq -0.54 (2.4p)$

$$\log_e 2 - \cos 2 \doteq 1.11 \text{ (2 dp)} \quad (1)$$

$\therefore$ There is a root between 1 & 2

b) If $f(x) = \log_e x - \cos x$

$$f'(x) = \frac{1}{x} + \sin x$$

$$\text{Now } a_2 = a_1 - \frac{f(a_1)}{f'(a_1)} \quad (3)$$

$$\div 1.2 = \left[\frac{-0.180036197}{1.765372419} \right]$$

$$\approx 1.30 \text{ (2 d.p.)}$$

(iii) If $x^2 = 607$, $y = \frac{12}{60}$

$$\therefore \frac{dy}{dx} = \frac{2x}{4a}$$

When $x = 2cp$

$$\frac{dy}{dx} = \frac{4ap}{4a}$$

$\therefore$ The grad of the normal is $-\frac{1}{p}$

∴ the eqⁿ is

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

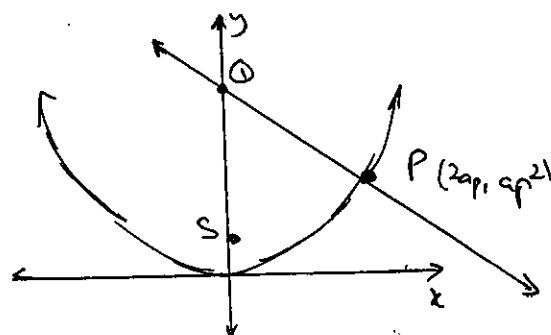
$$py - ap^3 = -x + 2ap$$

$$1 \leq x + py = 2ap + ap^3.$$

When $x = 0$, $y = 2ap + ap^3$
 $y = 2a + ap^2$.

$\therefore Q$ is the pt $(0, 2a + ap^2)$

ΔS is the pt $(0, a)$



$$\therefore QS = 2a + ap^2 - a$$
$$= a + ap^2$$

Area = $\frac{1}{2} \times b \times h$

$$= \frac{1}{2} (a + ap^2) \times 2ap$$

$$= a_p(a + a_p^2)$$

$$= a^2 p + a^2 p^3 \text{ units}^3$$

Question 6:

(i) $\sin 2\theta = \cos \theta$

$$\sin 2\theta - \cos \theta = 0$$

$$2 \sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (2 \sin \theta - 1) = 0$$

$$\therefore \cos \theta = 0 \quad \text{or} \quad \sin \theta = \frac{1}{2} \quad (2)$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\text{or } \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots$$

$$\text{i.e. } \theta = 2n\pi \pm \frac{\pi}{2} \quad \text{or} \quad n\pi + (-1)^n \cdot \frac{\pi}{6}$$

(ii) a) $\frac{d}{dx} \left[x \sin^{-1} x + \sqrt{1-x^2} \right]$

$$= \frac{d}{dx} \left[x \sin^{-1} x + (1-x^2)^{\frac{1}{2}} \right] \quad (2)$$

$$= \sin^{-1} x \cdot 1 + x \cdot \frac{1}{\sqrt{1-x^2}} + \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot -2x$$

$$= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1} x$$

b) $V = \pi \int_a^b y^2 dx$

$$= \pi \int_0^{\frac{1}{2}} \sin^{-1} x dx \quad (3)$$

$$= \pi \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^{\frac{1}{2}}$$

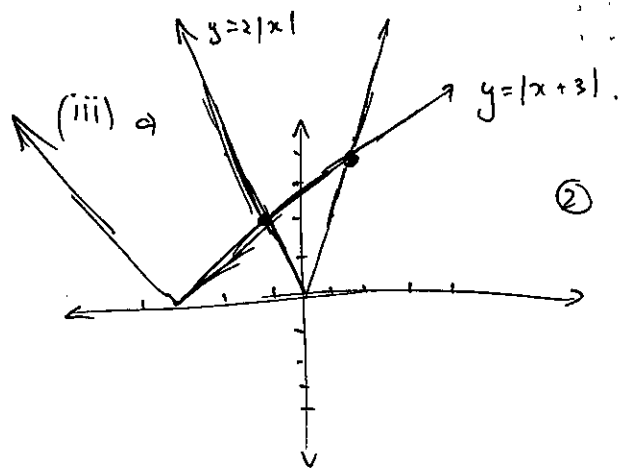
$$= \pi \left[\left(\frac{1}{2} \sin^{-1} \frac{1}{2} + \sqrt{1-\frac{1}{4}} \right) - (0 \cdot \sin^{-1} 0 + \sqrt{1-0}) \right]$$

$$= \pi \left[\left(\frac{1}{2} \cdot \frac{\pi}{6} + \frac{\sqrt{3}}{2} \right) - (0+1) \right]$$

$$= \pi \left(\frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \right)$$

$$= \frac{\pi^2}{12} + \pi \left(\frac{\sqrt{3}-2}{2} \right)$$

$$= \frac{\pi^2}{12} + \frac{\pi}{2} (\sqrt{3}-2) \text{ units}^3$$



b) If $2x = x+3$, If $-2x = x+3$

$$x = 3$$

$$-3x = 3$$

$$x = -1$$

$\therefore$ From the graph of

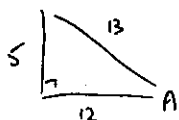
$$2|x| \leq |x+3|$$

$$\text{the solution: } -1 \leq x \leq 3$$

Question 7:

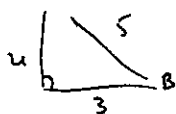
(i) Let $\sin^{-1} \frac{5}{13} = A$

$\therefore \sin A = \frac{5}{13}$



Let $\sin^{-1} \frac{4}{5} = B$

$\therefore \sin B = \frac{4}{5}$



$\therefore \tan \left[\sin^{-1} \frac{5}{13} - \sin^{-1} \frac{4}{5} \right]$

$= \tan [A - B]$

$= \frac{\tan A - \tan B}{1 + \tan A \tan B}$

$= \frac{\frac{5}{12} - \frac{4}{3}}{1 + \frac{5}{12} \times \frac{4}{3}}$

(3)

$= \frac{\frac{5}{12} - \frac{16}{12}}{1 + \frac{5}{9}}$

$= \frac{-\frac{11}{12}}{\frac{14}{9}}$

$= -\frac{11}{12} \times \frac{9}{14}$

$= -\frac{33}{56}$

(ii) If $x^2 = 8y$

$y = \frac{x^2}{8}$

$\frac{dy}{dx} = \frac{2x}{8}$

If $x = 4t$, $\frac{dy}{dx} = t$

If $x = 8t$, $\frac{dy}{dx} = 2t$

$\therefore$ Eqⁿ of tangent at P is

$y - 2t^2 = t(x - 4t)$

$y - 2t^2 = tx - 4t^2$

i.e. $y = tx - 2t^2$ --- (1)

$\therefore$ Eqⁿ of tangent at Q is

$y - 8t^2 = 2t(x - 8t)$

$y - 8t^2 = 2tx - 16t^2$

i.e. $y = 2tx - 8t^2$ --- (2)

$\therefore$ T is given by the pt of intersection of (1) & (2)

$\therefore tx - 2t^2 = 2tx - 8t^2$

$6t^2 = tx$

$x = 6t$

$y = 4t^2$

T is the pt $(6t, 4t^2)$

(4)

If $x = 6t$

$t = \frac{x}{6}$

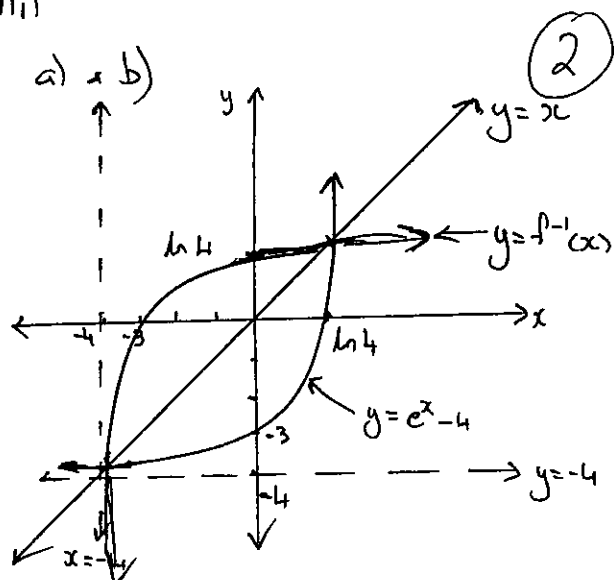
$\therefore$ if $y = 4t^2$

$y = 4 \times \left(\frac{x}{6}\right)^2$

$= \frac{4x^2}{36}$

i.e. $y = \frac{x^2}{9}$

(iii)



c) $f: y = e^x - 4$
is a one to one function
 $\therefore$ its inverse exists

$$f^{-1}: x = e^y - 4$$

$$\therefore e^y = x + 4$$

$$y = \log_e(x+4)$$

(1)

d) The points of intersection of
the two curves must lie
on the line $y = x$ since
they are inverses

(2)

$$\therefore e^x - 4 = x$$

$$\text{i.e. } e^x - 4 - x = 0.$$